First-Principles Three-Generation Standard Model Derivation in PSLT

Bo-Yu Chen[1]

[1] *Independent Researcher*

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We formulate the Projection Spectral Layer Theory (PSLT) as a first-principles projection-spectral derivation of the three-generation Standard Model. The central geometric input is a two-center projection divisor $R = p_+ + p_-$ on the compact projection surface $C = \mathbb{CP}^1$. The generation bundle is $L_R = \mathcal{O}(R) \simeq \mathcal{O}(2)$. By Riemann–Roch and Serre duality,

$$\dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3,$$

so the low-energy chiral Hilbert space contains exactly three copies of the anomaly-free Standard Model family representation and no fourth chiral generation. The same two-center geometry yields the conformal factor Ω through a variational Poisson constraint and induces the PSLT spectral operator $[-\nabla^2 + V_{\mathrm{eff}}]\psi_N = \omega_N^2 \psi_N$. The quantities g_N, Γ_N, B_N are defined as spectral degeneracy, two-center Schur-complement kinetics, and Weyl-normalized visible-sector overlap, respectively. The normalized probability

$$P_N(t) = \frac{B_N g_N (1 - \mathrm{e}^{-\Gamma_N t})}{\sum_K B_K g_K (1 - \mathrm{e}^{-\Gamma_K t})}$$

then follows as a conditional spectral-measure probability generated by the formation semigroup. Thus the three-generation structure is a topological consequence of the two-center projection geometry.

CONTENTS

I. CORE CLAIM

The central PSLT statement is the following:

$$\boxed{\mathcal{H}^{\text{4D}}_{\text{chiral}} = \mathcal{R}^{(1)}_{\text{SM}} \otimes H^0\big(\mathbb{CP}^1, \mathcal{O}(p_+ + p_-)\big)\,.} \tag{1}$$

Since p_+ and p_- are the two projection branch points,

$$\mathcal{O}(p_+ + p_-) \simeq \mathcal{O}(2), \qquad \dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3. \tag{2}$$

Therefore

$$\boxed{\mathcal{H}^{\text{4D}}_{\text{chiral}} = \mathcal{R}^{(1)}_{\text{SM}} \oplus \mathcal{R}^{(2)}_{\text{SM}} \oplus \mathcal{R}^{(3)}_{\text{SM}}.} \tag{3}$$

The probability law P_N describes formation and visibility phenomenology. The family count itself is fixed by the zero-mode dimension of the projection line bundle.

Remark 1. In this manuscript, "first principles" means consequences of the specified local gauge- and diffeomorphism-invariant projection theory in the topological sector $(C, R) = (\mathbb{CP}^1, p_+ + p_-)$. The theorem below is exact inside that parent theory. Changing C, adding projection branch points, or replacing L_R changes the topological sector and is a different theory.

II. PARENT PROJECTION THEORY

A. Fields and action

Let the spacetime be M_4, let the internal projection surface be

$$C \simeq \mathbb{CP}^1, \tag{4}$$

and let the two-center projection branch divisor be

$$R = p_+ + p_-. \tag{5}$$

The generation line bundle is

$$L_R = \mathcal{O}(R) \simeq \mathcal{O}(2). \tag{6}$$

The parent action is organized as

$$S_{\text{FP}} = S_{\text{grav}} + S_{\text{gauge}} + S_{\text{Higgs}} + S_{\text{fermion}} + S_{\Omega} + S_{\text{proj}} + S_{\text{vis}}. \tag{7}$$

Here the visible gauge group is

$$G_{\text{SM}} = SU(3)_c \times SU(2)_L \times U(1)_Y, \tag{8}$$

and the one-family representation $\mathcal{R}_{\mathrm{SM}}^{(1)}$ is fixed in Sec. III. The generation multiplicity is not inserted by copying this representation. Instead, the four-dimensional chiral fields arise from the Dolbeault zero modes on C:

$$\Psi_\alpha(x,z)=\sum_I \psi_{\alpha I}(x)s_I(z)+\text{massive nonchiral modes},\qquad s_I\in H^0(C,L_R),\tag{9}$$

where α labels the one-family SM representation.

The projection constraint may be written topologically as

$$S_{\mathrm{proj}}=\int_C \lambda_R\left[\frac{\mathrm{i}}{2\pi}F_{A_R}-\delta^{(2)}_{p_+}-\delta^{(2)}_{p_-}\right],\tag{10}$$

so that

$$c_1(L_R)=[p_+]+[p_-],\qquad \deg L_R=2.\tag{11}$$

Equation (10) is a compact way of saying that the two centers define the divisor of the generation bundle itself, not merely a scan ansatz.

B. Two-center conformal factor from a variational constraint

The two-center conformal factor is obtained from the action

$$S_\Omega=\frac{\kappa}{2}\int_{\mathbb{R}^3}\mathrm{d}^3x\,|\nabla\Omega|^2-4\pi\kappa\int_{\mathbb{R}^3}\mathrm{d}^3x\,\sigma(x)(\Omega-1),\tag{12}$$

with source

$$\sigma(x)=a\left[\rho_\varepsilon(x-x_+)+\rho_\varepsilon(x-x_-)\right].\tag{13}$$

Variation gives, up to boundary terms,

$$\delta S_\Omega=-\kappa\int_{\mathbb{R}^3}\mathrm{d}^3x\,\left(\nabla^2\Omega+4\pi\sigma\right)\delta\Omega.\tag{14}$$

Thus

$$\boxed{\nabla^2\Omega(x)=-4\pi\sigma(x).}\tag{15}$$

For the regularized Coulomb kernel

$$\rho_\varepsilon(r)=\frac{3\varepsilon^2}{4\pi(r^2+\varepsilon^2)^{5/2}},\qquad \int_{\mathbb{R}^3}\mathrm{d}^3x\,\rho_\varepsilon=1,\tag{16}$$

the asymptotically flat solution is

$$\boxed{\Omega(\rho,z;D)=1+a\left(\frac{1}{r_+}+\frac{1}{r_-}\right),\qquad r_\pm=\sqrt{\rho^2+(z\mp D/2)^2+\varepsilon^2}.}\tag{17}$$

This gives the two-center conformal factor as an action-derived constraint of the parent theory.

III. ONE STANDARD MODEL FAMILY FROM ANOMALY CANCELLATION

The topological argument fixes the number of copies. We first recall why the minimal one-family hypercharge pattern is fixed, up to overall normalization, by gauge invariance and anomaly cancellation.

Take the left-handed Weyl fields

$$Q:(3,2)_{Y_Q}, \quad u^c:(\bar{3},1)_{Y_u}, \quad d^c:(\bar{3},1)_{Y_d}, \qquad L:(1,2)_{Y_L}, \quad e^c:(1,1)_{Y_e}, \tag{18}$$

and a Higgs doublet

$$H:(1,2)_h. \tag{19}$$

Set $Y_Q = q$. Yukawa gauge invariance requires

$$QHu^c: \qquad q+h+Y_u=0, \qquad\qquad QH^\dagger d^c: \qquad q-h+Y_d=0, \tag{20}$$

$$LH^\dagger e^c: \qquad Y_L-h+Y_e=0. \tag{21}$$

Hence

$$Y_u=-q-h, \qquad Y_d=-q+h, \qquad Y_e=h-Y_L. \tag{22}$$

The $SU(3)^2U(1)$ anomaly gives

$$2Y_Q+Y_u+Y_d=0, \tag{23}$$

which is automatically satisfied by Eq. (22). The $SU(2)^2U(1)$ anomaly gives

$$3Y_Q+Y_L=0, \qquad Y_L=-3q. \tag{24}$$

The mixed gravitational anomaly gives

$$6Y_Q+3Y_u+3Y_d+2Y_L+Y_e=0. \tag{25}$$

Substituting Eqs. (22) and (24),

$$6q+3(-q-h)+3(-q+h)+2(-3q)+(h+3q)=0, \tag{26}$$

so

$$h=3q. \tag{27}$$

Therefore

$$Y_u=-4q, \qquad Y_d=2q, \qquad Y_L=-3q, \qquad Y_e=6q, \qquad Y_H=3q. \tag{28}$$

The $U(1)^3$ anomaly then cancels:

$$6q^3 + 3(-4q)^3 + 3(2q)^3 + 2(-3q)^3 + (6q)^3 = (6 - 192 + 24 - 54 + 216)q^3 = 0. \tag{29}$$

With the conventional normalization $q = 1/6$,

$$\boxed{(Y_Q, Y_u, Y_d, Y_L, Y_e, Y_H) = \left(\frac{1}{6}, -\frac{2}{3}, \frac{1}{3}, -\frac{1}{2}, 1, \frac{1}{2}\right).} \tag{30}$$

Adding a sterile $\nu^c : (1,1)_0$ does not affect these anomaly equations.

Proposition 1. *For the minimal chiral SM field content with the Yukawa couplings above, the one-family hypercharge pattern is fixed up to overall $U(1)_Y$ normalization.*

The number of such anomaly-free copies is not fixed by anomaly cancellation. That number is fixed by the projection geometry in the next section.

IV. THE THREE-GENERATION THEOREM

Theorem 1 (Two-center projection implies three generations)**.** *Let $C = \mathbb{CP}^1$, let $R = p_+ + p_-$, and let $L_R = \mathcal{O}(R)$. Then*

$$\dim H^0(C, L_R) = 3. \tag{31}$$

Consequently, the four-dimensional low-energy chiral SM spectrum contains exactly three generations:

$$\mathcal{H}^{\text{4D}}_{\text{chiral}} = \mathcal{R}^{(1)}_{\text{SM}} \oplus \mathcal{R}^{(2)}_{\text{SM}} \oplus \mathcal{R}^{(3)}_{\text{SM}}. \tag{32}$$

Proof. Riemann–Roch for a line bundle L on a compact Riemann surface C gives

$$h^0(C, L) - h^1(C, L) = \deg L + 1 - g(C), \tag{33}$$

where $h^i(C, L) = \dim H^i(C, L)$. For $C = \mathbb{CP}^1$, $g(C) = 0$. Because $R = p_+ + p_-$, $\deg L_R = 2$. Thus

$$h^0(C, L_R) - h^1(C, L_R) = 2 + 1 = 3. \tag{34}$$

By Serre duality,

$$h^1(C, L_R) = h^0(C, K_C \otimes L_R^{-1}). \tag{35}$$

On $\mathbb{CP}^1$, $K_C = \mathcal{O}(-2)$, while $L_R^{-1} = \mathcal{O}(-2)$. Hence

$$K_C \otimes L_R^{-1} \simeq \mathcal{O}(-4). \tag{36}$$

A negative-degree line bundle on $\mathbb{CP}^1$ has no nonzero holomorphic sections, so

$$h^0(\mathbb{CP}^1, \mathcal{O}(-4)) = 0, \qquad h^1(C, L_R) = 0. \tag{37}$$

Equation (34) therefore gives

$$h^0(C, L_R) = 3. \tag{38}$$

Tensoring these three zero modes with the one-family SM representation gives exactly three chiral SM families. $\square$

Equivalently, in index-theorem form,

$$\operatorname{ind}\bar{\partial}_{L_R} = \int_C \operatorname{ch}(L_R)\,\operatorname{td}(T_C) = \deg L_R + 1 - g(C) = 3. \tag{39}$$

The vanishing of h^1 upgrades the index to the actual zero-mode count.

A. Explicit zero-mode basis

Choose an affine coordinate z on $\mathbb{CP}^1$. A basis of $H^0(\mathbb{CP}^1, \mathcal{O}(2))$ is

$$s_0(z) = 1, \qquad s_1(z) = z, \qquad s_2(z) = z^2. \tag{40}$$

Thus the chiral field expansion is

$$\Psi_\alpha(x, z) = \sum_{I=0}^{2} \psi_{\alpha I}(x) s_I(z) + \sum_{\lambda>0} \Psi_{\alpha,\lambda}(x)\chi_\lambda(z). \tag{41}$$

The first sum is the protected chiral sector. The second sum contains massive nonchiral modes in

$$\operatorname{Ker}(\bar{\partial}_{L_R})^{\perp}. \tag{42}$$

They may exist as heavy excitations, but they are not additional SM generations.

B. Why there is no fourth generation

A fourth chiral SM generation would require a fourth independent holomorphic section

$$s_3 \in H^0(\mathbb{CP}^1, \mathcal{O}(2)), \qquad s_3 \notin \operatorname{span}\{1, z, z^2\}. \tag{43}$$

But Theorem 1 proves that this vector space has dimension exactly three. Therefore no fourth chiral SM generation exists in this topological sector. The only ways to obtain four chiral zero modes are to change the theory, for example by replacing $R = p_+ + p_-$ with a degree-three divisor or by changing the compact projection surface.

C. Exact numerical sanity check

The proof is topological, but the dimension count has a simple finite-dimensional check. Evaluate the three basis sections at $z = -1, 0, 1$:

$$V = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}. \tag{44}$$

Then

$$\det V = 2 \neq 0, \qquad \operatorname{rank} V = 3. \tag{45}$$

A direct floating-point singular-value check gives

$$\sigma(V) \simeq (2.13577921,\ 1.41421356,\ 0.66215345), \tag{46}$$

again confirming that the three holomorphic sections are independent. No fourth independent section can be tested because $\mathcal{O}(2)$ admits only quadratic homogeneous polynomials. This numerical check is not a substitute for Riemann–Roch; it is a transparent verification of the explicit basis in Eq. (40).

V. GEOMETRY-TO-SPECTRUM MAP

The three-generation theorem is topological and does not depend on the scalar scan. The scalar spectral chain remains important because it supplies the PSLT formation rates and visible overlap profiles.

Take a locally conformally flat metric

$$g_{\mu\nu} = \Omega^2(x)\eta_{\mu\nu}. \tag{47}$$

For a curvature-coupled scalar,

$$(\Box_g - m_0^2 - \xi R)\Phi = 0. \tag{48}$$

Set

$$\Phi(t, x) = \mathrm{e}^{-\mathrm{i}\omega t}\Omega^{-1}(x)\psi(x). \tag{49}$$

Using the conformal identity

$$\left(\Box_g - \frac{1}{6}R\right)(\Omega^{-1}\psi) = \Omega^{-3}\Box_\eta\psi \tag{50}$$

and

$$R = -6\Omega^{-3}\nabla^2\Omega, \tag{51}$$

one obtains

$$\left[-\nabla^2 + m_0^2\Omega^2 + (1-6\xi)\Omega^{-1}\nabla^2\Omega\right]\psi = \omega^2\psi. \tag{52}$$

Define

$$V_{\text{eff}}(x) = m_0^2\Omega^2 + (1-6\xi)\Omega^{-1}\nabla^2\Omega. \tag{53}$$

Then the geometry-to-spectrum map is

$$\boxed{[-\nabla^2 + V_{\text{eff}}(x)]\psi_N = \omega_N^2\psi_N.} \tag{54}$$

Together with Eq. (17), this gives the PSLT chain

$$\Omega \longrightarrow V_{\text{eff}} \longrightarrow \{\omega_N, \psi_N\}. \tag{55}$$

VI. IDENTIFICATION OF PSLT LAYERS

The PSLT layer labels in the chiral sector are identified with an orthonormal basis of the topological zero modes:

$$N = 1, 2, 3 \quad \Longleftrightarrow \quad u_N(z) = \mathrm{GramSchmidt}\{1, z, z^2\}_N \in H^0(\mathbb{CP}^1, \mathcal{O}(2)). \tag{56}$$

The chiral four-dimensional field is

$$\Psi_\alpha(x, z) = \sum_{N=1}^{3} \psi_{\alpha N}(x) u_N(z) + \text{massive nonchiral modes}. \tag{57}$$

Thus

$$\boxed{\text{PSLT layers } N = 1, 2, 3 \text{ are the three topologically protected SM-family zero modes.}} \tag{58}$$

The scalar spectrum in Eq. (54) controls formation and visibility of these layers, but it does not create or remove the chiral family index.

VII. FIRST-PRINCIPLES DEFINITIONS OF g_N, Γ_N, B_N

A. Spectral degeneracy g_N

Let Π_N be the spectral projector onto the N-th chiral zero-mode bin. Define

$$g_N^{\text{chiral}} = \mathrm{Tr}_{\mathcal{H}_{\text{gen}}} \Pi_N, \qquad \mathcal{H}_{\text{gen}} = H^0(\mathbb{CP}^1, \mathcal{O}(2)). \tag{59}$$

For the three protected zero modes,

$$\boxed{g_1 = g_2 = g_3 = 1, \qquad g^{\mathrm{chiral}}_{N\geq 4} = 0.} \tag{60}$$

If heavy nonchiral excitations are included, their degeneracy is a separate spectral trace, for example

$$g^{\mathrm{heavy}}_{\lambda}(\Lambda_{\mathrm{IR}}) = \mathrm{Tr}_{\mathcal{H}_{\mathrm{heavy}}}\left[\Pi_\lambda \Theta\Big(\Lambda^2_{\mathrm{IR}} - \mathcal{D}^\dagger_C \mathcal{D}_C\Big)\right]. \tag{61}$$

Such states do not modify $N_{\mathrm{gen}} = 3$.

B. Two-center Schur-complement kinetics Γ_N

For each protected layer $N = 1, 2, 3$, the two projection centers define a two-dimensional collective-coordinate subspace

$$\mathcal{C}_N = \mathrm{span}\{|N,+\rangle,\ |N,-\rangle\}. \tag{62}$$

Let P_N project onto $\mathcal{C}_N$, $Q_N = 1 - P_N$, and let $\mathcal{K}_N$ be the Euclidean fluctuation operator obtained from the parent action. The Feshbach–Schur effective operator is

$$M_N = P_N \mathcal{K}_N P_N - P_N \mathcal{K}_N Q_N (Q_N \mathcal{K}_N Q_N)^{-1} Q_N \mathcal{K}_N P_N. \tag{63}$$

In the two-center basis it has the form

$$M_N = \begin{pmatrix} \gamma_{N,+} & \chi_N \sqrt{\gamma_{N,+}\gamma_{N,-}} \\ \chi_N \sqrt{\gamma_{N,+}\gamma_{N,-}} & \gamma_{N,-} \end{pmatrix}. \tag{64}$$

The diagonal channel factors are WKB-suppressed:

$$\gamma_{N,\pm} = A_{N,\pm} \exp[-2S^{\mathrm{WKB}}_{N,\pm}], \qquad S^{\mathrm{WKB}}_{N,\pm} = \int_{\mathcal{B}_\pm} \mathrm{d}s\, \sqrt{V_{\mathrm{eff}}(s) - \omega^2_N}. \tag{65}$$

The formation rate is

$$\boxed{\Gamma_N(D,\eta) = \max\{0, \lambda_+(M_N(D,\eta))\},} \tag{66}$$

where

$$\lambda_+(M_N) = \frac{\gamma_{N,+} + \gamma_{N,-}}{2} + \frac{1}{2}\sqrt{(\gamma_{N,+} - \gamma_{N,-})^2 + 4\chi^2_N \gamma_{N,+}\gamma_{N,-}}. \tag{67}$$

The rank two structure is therefore a consequence of the two-center collective-coordinate reduction, not a phenomenological closure assumption.

C. Weyl-normalized visibility B_N

For the conformal Dirac and Higgs fields,

$$\Psi = \Omega^{-3/2}\psi, \qquad H = \Omega^{-1}\phi. \tag{68}$$

The Yukawa density obeys

$$\sqrt{-g}\,\bar{\Psi}H\Psi \propto \Omega^4 \cdot \Omega^{-3/2} \cdot \Omega^{-1} \cdot \Omega^{-3/2} = \Omega^0. \tag{69}$$

Thus the minimal conformal construction has no additional Ω-prefactor in the overlap kernel. For a visible-sector operator $\mathcal{O}_{\text{vis}}$, define

$$\boxed{B_N = \frac{\langle u_N|\mathcal{O}_{\text{vis}}^{\dagger}\mathcal{O}_{\text{vis}}|u_N\rangle}{\sum_{I=1}^{3}\langle u_I|\mathcal{O}_{\text{vis}}^{\dagger}\mathcal{O}_{\text{vis}}|u_I\rangle}} \tag{70}$$

or, in Yukawa-overlap form,

$$y_N^{\text{eff}}(D) = y_0 \int 2\pi\rho\,\mathrm{d}\rho\,\mathrm{d}z\,u_N(\rho, z; D) f_L(\rho, z) f_R(\rho, z) W_{\text{frame}}, \tag{71}$$

$$B_N = \frac{|y_N^{\text{eff}}|^2}{\sum_{I=1}^{3}|y_I^{\text{eff}}|^2}. \tag{72}$$

Thus B_N is a normalized overlap of the same three topological zero modes with the visible sector.

VIII. FORMATION PROBABILITY

For each chiral layer N, take the effective scalar formation law

$$\frac{\mathrm{d}p_N}{\mathrm{d}t} = \Gamma_N(1 - p_N), \qquad p_N(0) = 0. \tag{73}$$

The solution is

$$p_N(t) = 1 - \mathrm{e}^{-\Gamma_N t}. \tag{74}$$

If the N-th layer has degeneracy g_N and visible weight B_N, its unnormalized observed formation weight is

$$W_N(t) = B_N g_N \left(1 - \mathrm{e}^{-\Gamma_N t}\right). \tag{75}$$

Conditioning on the event that a chiral layer is observed gives

$$\boxed{P_N(t; D, \eta) = \frac{B_N g_N \left(1 - \mathrm{e}^{-\Gamma_N(D,\eta)t}\right)}{\sum_{K=1}^{3} B_K g_K \left(1 - \mathrm{e}^{-\Gamma_K(D,\eta)t}\right)}.} \tag{76}$$

Because $g_{N\geq 4}^{\text{chiral}} = 0$, the same expression may be written with a formal sum over all K in the chiral sector. Heavy nonchiral modes, if included in phenomenological spectra, must be treated as massive excitations rather than additional SM families.

IX. MAIN THEOREM

Theorem 2 (PSLT first-principles three-generation theorem). *Consider the local gauge- and diffeomorphism-invariant projection theory on $M_4 \times C$, with $C = \mathbb{CP}^1$, two projection branch points p_+, p_-, generation bundle $L_R = \mathcal{O}(p_+ + p_-)$, and visible matter content fixed by SM gauge invariance, Yukawa gauge invariance, and anomaly cancellation. Then:*

1. *The one-family SM hypercharge pattern is fixed up to normalization as*

$$(Y_Q, Y_u, Y_d, Y_L, Y_e, Y_H) = \left(\frac{1}{6}, -\frac{2}{3}, \frac{1}{3}, -\frac{1}{2}, 1, \frac{1}{2}\right).$$

2. *The number of low-energy chiral SM families is*

$$N_{\text{gen}} = \dim H^0(\mathbb{CP}^1, \mathcal{O}(p_+ + p_-)) = 3.$$

3. *There is no fourth chiral SM generation in this topological sector because*

$$\dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3.$$

4. *The two-center conformal factor follows from the variational equation $\nabla^2\Omega = -4\pi\sigma$ and generates the spectral operator $[-\nabla^2 + V_{\text{eff}}]\psi_N = \omega_N^2\psi_N$.*

5. *The PSLT probability law*

$$P_N = \frac{B_N g_N (1 - \mathrm{e}^{-\Gamma_N t})}{\sum_K B_K g_K (1 - \mathrm{e}^{-\Gamma_K t})}$$

is the conditional spectral-measure probability of the formation semigroup, with g_N from zero-mode degeneracy, Γ_N from the two-center Schur complement, and B_N from Weyl-normalized visible overlap.

The observed three-generation structure is therefore not imposed as a family-count input. It is the zero-mode dimension of the two-center projection line bundle.

X. PHENOMENOLOGICAL USE

The topological theorem fixes the chiral family count. Phenomenology then concerns how the three protected modes are formed, weighted, and observed. The scalar spectrum and overlap maps should be used for:

1. formation-rate predictions through $\Gamma_N(D, \eta)$;

2. visible-sector hierarchy through $B_N(D)$;

3. mass and Yukawa hierarchy modeling through y_N^{eff};

4. observable diagnostics such as $H \to ee$, $H \to \mu\mu$, and $H \to \tau\tau$.

These diagnostics may constrain the geometry parameters (D, η), but they do not determine the number of SM generations. That number is fixed by the degree-two projection divisor.

The clean separation is

$$\boxed{\begin{array}{c}\text{family count} = \dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3, \\ \text{phenomenology} = (g_N, \Gamma_N, B_N, P_N) \text{ on these modes.}\end{array}} \tag{77}$$

XI. CONCLUSION

The PSLT construction gives the family count as a topological theorem. Two projection centers define the divisor $R = p_+ + p_-$ on $C = \mathbb{CP}^1$. The associated line bundle $L_R = \mathcal{O}(R) = \mathcal{O}(2)$ has exactly three holomorphic sections. Tensoring those sections with the unique minimal anomaly-free SM family representation gives exactly three chiral SM generations.

The formation probability P_N describes formation and visibility of the three protected zero modes. The quantities g_N, Γ_N, B_N are spectral degeneracy, two-center Schur-complement kinetics, and Weyl-normalized visible overlap derived from the same parent projection theory. The essential statement is therefore:

$$\boxed{\begin{array}{c}\text{Three generations are } H^0 \text{ zero modes} \\ \text{of the two-center projection divisor.}\end{array}} \tag{78}$$

Appendix A: Relation to Earlier Draft Structure

This appendix records the correspondence between the earlier draft organization and the first-principles formulation used in the main text. It is included only as an editorial migration note and is not part of the logical derivation of the family count.

Remark 2. For future roadmap synchronization, the governing entry is: "Family count is topological: $N_{\mathrm{gen}} = \dim H^0(\mathbb{CP}^1, \mathcal{O}(p_+ + p_-)) = 3$. Do not use scan-level three-layer dominance or finite-volume no-fourth-layer certificates as the proof of the SM family count."

TABLE I. Editorial migration map from earlier draft modules to the present first-principles formulation.

Earlier draft module or wording	**Present formulation**
Two-center Ω as a projected-Poisson Green background	Ω follows from S_Ω by variation, giving $\nabla^2\Omega = -4\pi\sigma$.
Three-layer dominance inferred from a finite scan	Three generations follow from $\dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3$.
No-fourth-layer finite-volume Sturm/Sylvester certificate	No-fourth-generation follows from Riemann–Roch plus Serre duality. Finite-volume certificates may still test massive spectral thresholds, but not the chiral family count.
g_N as bounded phase-space baseline	g_N is the trace of the chiral spectral projector; $g_1 = g_2 = g_3 = 1$, $g_{N\geq 4}^{\mathrm{chiral}} = 0$.
Γ_N as rank-2 kinetic closure	$\Gamma_N = \max(0, \lambda_+(M_N))$, with M_N obtained from the two-center Feshbach–Schur complement.
B_N as EFT-normalized visibility factor	B_N is a Weyl-normalized Yukawa or EFT overlap of the same three topological zero modes.
P_N as master-equation baseline	P_N is the conditional probability generated by the formation semigroup.
$H \to \mu\mu$ mapping as support for generation selection	$H \to \mu\mu$ is an observable diagnostic of visibility and normalization, not the source of $N_{\mathrm{gen}} = 3$.

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